MARITIME POWER RESEARCH · CYBER DEFENSE · STRATEGIC STUDIES

Cyber power as a structural variable of maritime power

A normalized multiplicative model for multidomain assessment of maritime power

$$MP_{norm} = [(\ MI \times NP\ ) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/n_{eff}}$$

$n_{eff} = 3 + \gamma + \alpha$ · $MP_{norm} \in (0, 1]$ · weighted geometric mean (OECD–JRC, 2008; UNDP, 2010)

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Manuscript under review — IEEE Access (revised version, October 2026) · diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html

THE RESEARCH PROBLEM

A sound formula with a structural gap

THE CLASSICAL FORMULATION

$$MP = MI \times NP$$

Maritime power = maritime interests × naval power

Solís Oyarzún (1997) – Latin American systematization
Till (2007) – system of interconnected elements
Uribe Cáceres et al. (2016) – adoption for Colombia
Naval Development Plan 2042 (ARC, 2022) – institutional currency

THE CONCEPTUAL GAP

Cyberspace does not appear as a structural domain of maritime power

Navigation, communications, traffic management, ports and surveillance depend on interconnected digital infrastructures whose vulnerability can affect the exercise of maritime power without physical confrontation.

Nor is strategic will formalized as a variable of the system, nor is maritime security disaggregated.

The article updates the formula — it does not replace it — in continuity with the tradition of the field.

EMPIRICAL EVIDENCE

Cyberspace already degrades maritime power

USD 300 M

Cyberattack on Maersk (2017)

Operations disrupted and losses estimated at USD 300 million.

Wesley et al. (2019); Abbatemarco et al. (2024)

+900 %

Breaches in operational technology

A single compromised system can give access to the other onboard systems.

Akpan et al. (2022)

AIS · ECDIS

Critical maritime systems

Documented vulnerabilities in automatic identification and electronic charts.

Martínez et al. (2024)

2017 → 2021

Resolution MSC.428(98) – IMO

Cyber risk must be addressed in safety management systems.

IMO (2017)

Colombia: ~988,000 km² of jurisdictional maritime territory and eighteen recognized maritime interests (ARC, 2020, 2022; CCO, 2021).

What the article answers

RESEARCH QUESTION

How should the conceptual expression of maritime power be reformulated to incorporate cyberspace as a strategic domain?

HYPOTHESIS

Cyber power is a structural variable whose omission yields an incomplete representation of state capabilities and vulnerabilities.

OBJECTIVE

To develop an extended formulation that integrates cyber power, strategic will and maritime security, with international theoretical scope and an analytical illustration for Colombia.

THEORETICAL FRAMEWORK

A 130-year tradition, and its next link

1890
Mahan
Six conditions of sea power

1911
Corbett
Control of sea communications

1997
Solís Oyarzún
MP = MI × NP

2007
Till
System of interconnected elements

2016
Uribe Cáceres et al.
Adoption for Colombia; will and maritime consciousness

2026
This proposal
Extended, normalized formula

THREE LITERATURES THE ARTICLE BRINGS TOGETHER

Maritime power
Mahan · Corbett · Castex · Till · Solís Oyarzún · Uribe Cáceres

Maritime security
Bueger & Edmunds (2024) – IC · TE · BC

Cyber power
Nye (2010) · Bebber (2022) · Cabuya-Padilla et al. (2024, 2025)

STATE OF THE ART

Four gaps no antecedent resolves

1 **Cyber power is not formalized**

None of the 812 screened records formalizes it as a structural variable of a maritime power measure.

2 **Strategic will stays outside the system**

It is treated as a driver of naval power, not as a condition of the whole system.

3 **Maritime security not disaggregated**

Without the threat dimensions of the contemporary agenda or cyberspace.

4 **All existing measures are additive**

A deficit in one component is fully offset by a surplus in another.

Documented search (OpenAlex + Scopus, 13-oct-2026, 2014–2026): 1,045 records → 812 unique → 13 measure maritime or naval power → **0 include cyber power**. IMEMO, Chairil et al., Crisher and Souva, Zhang et al., Vargas-Suárez et al.: all additive.

METHOD

Analytical-conceptual approach, in four stages

Conceptual formalization precedes quantitative operationalization – the same tradition as Mahan, Corbett, Till and Solís Oyarzún.

1

STAGE 1

Documented search

Four queries in OpenAlex and Scopus; records and screening published.

2

STAGE 2

Critical analysis

Content and limits of the classical formula in its sources.

3

STAGE 3

Building the formula

Two levels; justified functional forms; normalization by n_{eff}.

4

STAGE 4

Illustration: Colombia

MS from Vargas-Suárez et al. (2021); other values illustrative.

Verification ≠ calibration ≠ external validation (Sargent, 2011). The model is verified; calibration and external validation are left as an agenda, with a stated protocol.

CENTRAL RESULT · LEVEL 1

The extended formula of maritime power

$$MP_{norm} = [(\ MI \times NP\ ) \times SW \times MS^{\gamma} \times CP^{\alpha}\ ]^{1/n_{eff}}$$

$$n_{eff} = 3 + \gamma + \alpha \rightarrow MP_{norm} \in (0,\ 1] \cdot \omega_{MI} = \omega_{NP} = \omega_{SW} = 1/n_{eff},\ \omega_{MS} = \gamma/n_{eff},\ \omega_{CP} = \alpha/n_{eff}$$

MI
Maritime interests
Purpose: what the state seeks at sea

NP
Naval power
Instrument: force, position and MDA

SW
Strategic will
Multiplier of the whole system

MS
Maritime security
Condition of the environment; weight γ

CP
Cyber power
Structural variable; weight α

Keeps the classical core. **Limited substitutability**: a deficit is only partly offset, and the collapse of any component cannot be offset (OECD–JRC, 2008; Greco et al., 2019).

Disaggregation of the five components

MI

$$MI = \Sigma\, w_i \cdot MI_i$$

Linear weighted sub-index: interests are substitutable and weights act as substitution rates. Colombia: n = 18 (CCO, 2021).

NP

$$NP = F \times P \times MDA^{\mu}$$

Force (with tactical-operational cyber defense), position and maritime domain awareness, raised to the level of F × P.

SW

$$SW = (PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3})^{1/\Sigma w}$$

Political, institutional and social will; geometric mean: none offsets the absence of another.

MS

$$MS = (IC^{a_1} \cdot TE^{a_2} \cdot BC^{a_3})^{1/\Sigma a}$$

Interstate conflicts, terrorism and blue crime: measures the incidence of threats, not capabilities.

CP

$$CP = \min\{1,\ (\beta_1 CS_m + \beta_2 CD_m + \beta_3 RC_m)(1 + \delta \cdot CDip)\}$$

Cybersecurity, strategic cyber defense and recovery capacity, amplified by cyber diplomacy.

Explicit operational boundaries: each indicator is assigned to a single component, to avoid double counting.

Cyber power, inside

CS_m **Maritime cybersecurity** – Controls, threat analysis and information sharing on the IT/OT assets of the domain.

CD_m **Strategic maritime cyber defense** – National, non-executional capacity: governance, cooperation, doctrine, science, education and deterrent posture.

RC_m **Maritime recovery capacity** – Speed of restoration: $RC_m = \min\{1, t_{ref}/t\}$.

CDip **Cyber diplomacy** – Amplifier only: never reduces CP; saturation keeps CP ≤ 1.

α

WEIGHT OF CYBER DEPENDENCE

Proposition 1: $MP_{norm} = G^{1-w} \cdot CP^{w}$, $w = \alpha/n_{eff}$.

Raising CP always raises the index; raising α raises it only if CP > G. Concavity holds for **any** α > 0.

Colombia: G = 0.464 > CP = 0.40 → a larger α lowers the index (0.464 → 0.450).

Cyber maturity is read on CP itself, in five bands: incipient, basic, consolidating, advanced, consolidated.

RESULTS · INTERPRETATION

A five-level analytical scale

High	**0.81 – 1.00**	Consolidated multidomain projection
Significant	**0.61 – 0.80**	Regional power with developed capabilities
Moderate	**0.41 – 0.60**	Developing power with identifiable gaps ★ **Colombia: MP_{norm} = 0.455**
Low	**0.21 – 0.40**	Incipient capability with structural vulnerabilities
Critical	**0.00 – 0.20**	Structural maritime vulnerability

FOUR USES

- Diagnosis of the current level
- Target level for planning
- Comparison among middle powers
- Scenarios in the simulator

Five equal intervals of the normalized range: conventions of the article, not thresholds estimated from data.

MODEL VERIFICATION

Three verification stages

A — Logical-mathematical properties

Collapse, monotonicity, concavity for any $\alpha, \gamma > 0$, bounds. 300,000 draws; maximum deviation 1.2×10^{-10}.

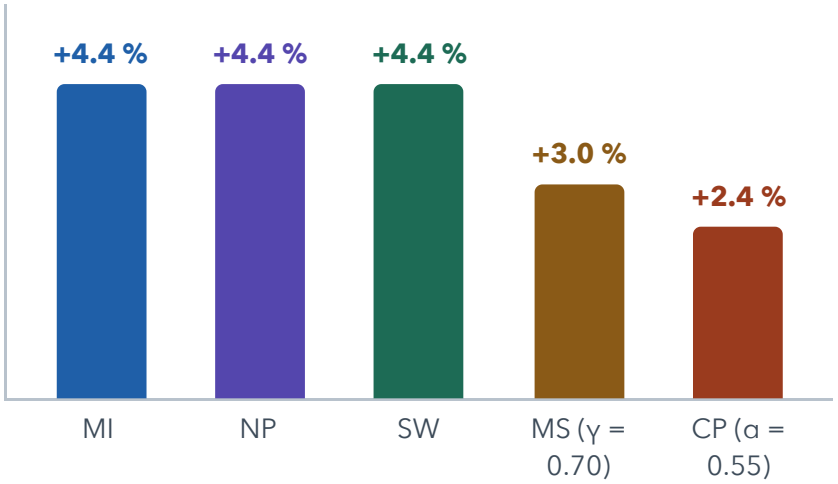
B — Sensitivity (+20%)

The elasticity of each component equals its normalized weight ω_k.

C — Constructed profiles

Critical ≈ 0.19 · reference ≈ 0.52 · consolidated ≈ 0.85: discriminates along the scale.

ΔMP_{norm} for a 20% improvement



Cyber power has the lowest proportional response ($\omega_{CP} = 0.129$). Sensitivity is not efficiency: comparing investments requires costs.

How robust is the result for Colombia?

	Components ± 0.10	Components ± 0.20
Median MP_{norm}	0.464	0.452
P5–P95 band	[0.406, 0.524]	[0.346, 0.557]
Moderate / Low	96.6 % / 3.4 %	78.1 % / 21.0 %
S_{Ti} of MS · γ	0.204 · 0.361	0.325 · 0.163
S_{Ti} of α	0.035	0.025
MS is the lowest	98.0 %	76.8 %

Monte Carlo with Sobol indices (Saisana et al., 2005; Saltelli et al., 2010); α and γ in [0.10, 1]. Ranges are the authors' assumptions, not estimates.

1 · Moderate is the most likely level

Robust to moderate uncertainty; with wide ranges, Low appears in 21% of the draws.

2 · MS and γ dominate the variance

Measuring maritime security and calibrating γ are the priorities of the index; α is the least influential factor.

3 · The aggregation rule does not change the level

Arithmetic mean 0.454 · unweighted geometric 0.433 · MP_{norm} 0.455: all three Moderate.

ANALYTICAL ILLUSTRATION

Colombia: what the model decomposes

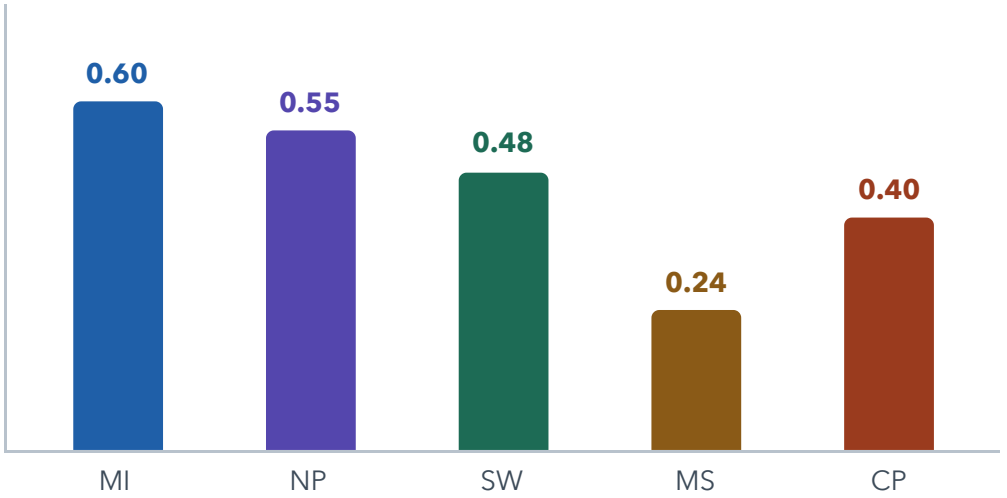
MP_{norm}

0.455

MODERATE LEVEL

Only MS comes from an external source: rank 75 of 99 in maritime security (Vargas-Suárez et al., 2021), converted into a position percentile, (99 − 75)/98 ≈ 0.24. The other values and the parameters are illustrative.

Reference values by component (α = 0.55; γ = 0.70)



Maritime security is the lowest component and the one with the largest marginal return; cyber power is the second lowest, at the top of the *basic* band.

An illustration, not a validation: the value is not compared with the Vargas-Suárez et al. index, which is on another scale.

STRATEGIC IMPLICATION

Is cyber power the most efficient investment vector? It depends on costs

Absolute marginal return $\partial MP_{norm}/\partial X_k$

Component	Return
MS – Maritime security	0.312
SW – Strategic will	0.223
NP – Naval power	0.195
MI – Maritime interests	0.178
CP – Cyber power	0.147

With the Colombian values, cyber power ranks last in marginal return.

Corollary 1 – cost condition

$$c_k/c_{CP} > (\omega_k/X_k) \,/\, (\omega_{CP}/CP)$$

Cyber power would be the most efficient vector only if raising each other component by one unit cost at least:

MI	NP	SW	MS
1.21×	1.32×	1.52×	2.12×

what it costs to raise cyber power. The model contains no costs: this is a testable condition, not a result.

The submitted version claimed cyber power was "the most efficient vector"; the revision withdrew that claim and replaced it with this condition.

Five contributions

1	**Cyber power as a structural variable**	With explicit operational boundaries and α as a weight, not a capability.
2	**Strategic will as a multiplier of the system**	It conditions all of maritime power, not only naval power.
3	**Disaggregated maritime security**	IC · TE · BC as incidence of threats, with weight γ.
4	**Normalization, scale and verification**	$MP_{norm} \in (0, 1]$, five levels, Proposition 1 and Corollary 1; verification distinguished from calibration and validation.
5	**MDA ≠ maritime consciousness**	Technical-operational capability in NP; cultural construct in SW.

DISCUSSION

Stated limitations, an agenda with a protocol

LIMITATIONS

- Verified, but neither calibrated nor externally validated.
- In the illustration, only MS comes from an external source.
- Static model; the enabling effect of cyber power on NP and MS is argued but not represented.
- Robust to moderate uncertainty, not to large uncertainty.

CALIBRATION AND VALIDATION PROTOCOL (NATIONAL INDEX)

1. **Expert calibration:** AHP panel on the five components; γ and α as weight ratios.
2. **Independent inputs:** published sources (ITU, 2024; NCSI; Wild and Fraser, 2026).
3. **Uncertainty:** Monte Carlo with each expert's weights and each source's error.
4. **Comparison:** rank correlation with Vargas-Suárez et al. (2021), with the criterion fixed beforehand.

Also: regional extension and dynamic modeling with SERDUX-MARCIM.

CONCLUSIONS

Maritime power is also exercised in cyberspace

01 The classical formula remains valid but is no longer sufficient: the proposal updates it while keeping its core.

02 Cyber power, strategic will and maritime security enter as structural variables, in a non-compensatory normalized multiplicative form.

03 Colombia is illustrated at the Moderate level ($MP_{norm} \approx 0.455$), with maritime security as the largest gap; this is not a definitive measurement.

04 Whether cyber power is the best investment depends on relative costs (Corollary 1); the national index is the next stage.

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Open-access interactive simulator: diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html

Thank you